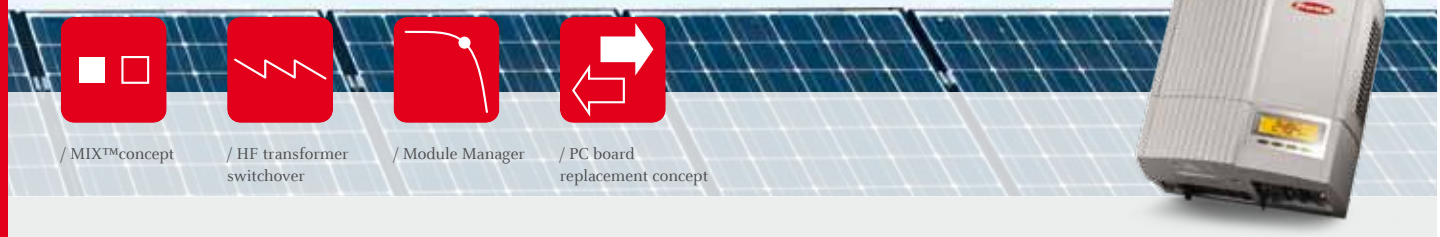
SHIFTING THE LIMITS

FRONIUS IG TRANSFORMER INVERTER

/ Updated design for Australian conditions.



/ MIX™concept

/ HF transformer switchover

/ Module Manager

/ PC board replacement concept

/ With the Fronius IG product family, Fronius has launched a generation of inverters rated from 1.5 to 5 kW that is compatible with all solar modules. What makes the inverters so appealing is their intuitive operation and ease of use, together with their highly informative analyses of system values in every situation. In short: a PV inverter that any system operator would welcome.

TECHNICAL DATA: FRONIUS IG

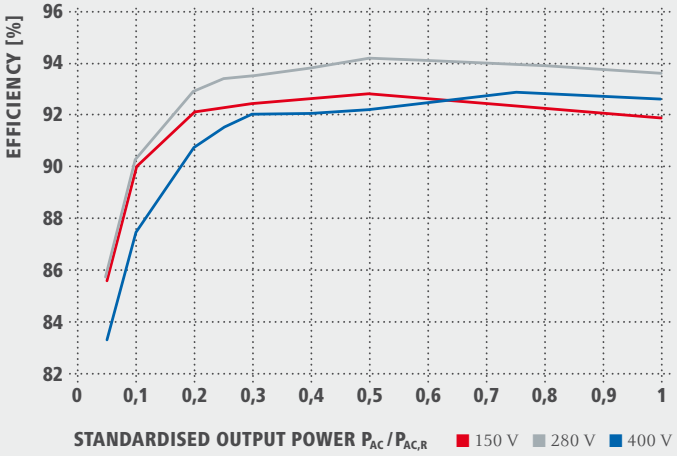
INPUT DATA	IG 15	IG 20	IG 30	IG 40	IG 50[1]	IG 60 HV
DC maximum power at cos φ = 1	1,610 W	2,150 W	2,850 W	4,410 W	4,950 W	5,380 W
Max. input current ($I_{dc\ max}$)	10.8 A	14.3 A	19.0 A	29.4 A	33.0 A	35.8 A
Min. input voltage ($U_{dc\ min}$)	150 V					
Feed-in start voltage ($U_{dc\ start}$)	170 V					
Nominal input voltage ($U_{dc,r}$)	280 V					
Max. input voltage ($U_{dc\ max}$)	500 V				530 V	
MPP voltage range ($U_{mpp\ min}$ – $U_{mpp\ max}$)	150 V – 400 V					
Number of DC inputs	5					

OUTPUT DATA	IG 15	IG 20	IG 30	IG 40	IG 50[1]	IG 60 HV
AC nominal output ($P_{ac,r}$)	1,300 W	1,800 W	2,500 W	3,500 W	4,600 W	4,600 W
Max. output power	1,500 W	2,000 W	2,650 W	4,100 W	4,600 W	5,000 W
Max. output current ($I_{ac\ max}$)	6.5 A	8.7 A	11.5 A	17.8 A	20.0 A	21.7 A
Grid connection ($U_{ac,r}$)	1-NPE 230 V					
Min. output voltage ($U_{ac\ min}$)	180 V					
Max. output voltage ($U_{ac\ max}$)	270 V					
Frequency (f_r)	50 Hz / 60 Hz					
Frequency range (f_{min} – f_{max})	47 Hz – 65 Hz					
Distortion factor	< 3 %					
Power factor ($\cos \varphi_{ac,r}$)	1					

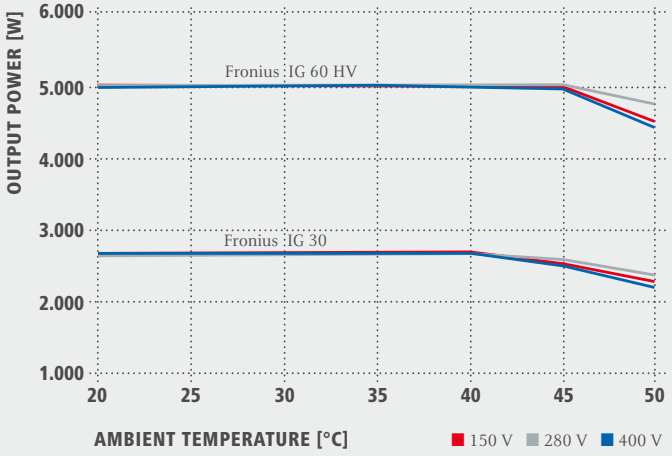
GENERAL DATA	IG 15	IG 20	IG 30	IG 40	IG 50[1]	IG 60 HV
Dimensions (height x width x depth)	366 x 344 x 220 mm / 500 x 435 x 225 mm[2]			610 x 344 x 220 mm / 733 x 435 x225 mm[2]		
Weight	9 kg / 12 kg[2]			16 kg / 20 kg[2]		
Degree of protection	IP 21 / IP 45[2]					
Protection class	1					
Overvoltage category (DC / AC)	2 / 3					
Night-time consumption	< 1 W					
Inverter concept	HF transformer					
Cooling	Regulated air cooling					
Installation	Indoor and outdoor installation[2]					
Ambient temperature range	from -20°C – +50°C					
Permitted humidity	0% – 95%					
DC connection technology	Screw terminal connection 1.5 – 10 mm²; DC plug optional[3]					
AC connection technology	Screw terminal connection 1.5 – 16 mm²					
Certificates and compliance with standards	DIN V VDE V 0126-1-1, ÖVE/ÖNORM E 8001-4-712, UTE C15-712-1, EN 50438, G83, G59, C 10 / 11, CER 06-190, Guida per le connessioni alla rete elettrica di ENEL Distribuzione, AS 4777-1, AS 4777-2, AS 4777-3					

[1] Fronius IG 50 devices may only be used in Germany. [2] This applies to Fronius IG Outdoor [3] MC3, MC4 or Tyco

FRONIUS IG 60 HV EFFICIENCY CURVE



FRONIUS IG TEMPERATURE DERATING



TECHNICAL DATA: FRONIUS IG

EFFICIENCY	IG 15	IG 20	IG 30	IG 40	IG 50[1)]	IG 60 HV
Max. efficiency	94.2 %	94.3 %	94.3 %	94.3 %	94.3 %	94.3 %
European efficiency (η_{EU})	91.4 %	92.3 %	92.9 %	93.2 %	93.5 %	93.5 %
η at 5% $P_{ac,r}$[4)]	75.0 / 76.9 / 71.1 %	77.4 / 80.6 / 71.1 %	81.6 / 83.1 / 81.4 %	82.7 / 83.3 / 80.2 %	85.6 / 85.8 / 83.3 %	85.6 / 85.8 / 83.3 %
η at 10% $P_{ac,r}$[4)]	81.6 / 83.1 / 81.4 %	84.9 / 86.2 / 83.4 %	87.4 / 88.6 / 85.9 %	88.5 / 89.3 / 85.0 %	90.0 / 90.3 / 87.5 %	90.0 / 90.3 / 87.5 %
η at 20% $P_{ac,r}$[4)]	87.8 / 89.2 / 85.9 %	89.7 / 90.5 / 87.3 %	91.2 / 91.8 / 89.1 %	91.5 / 92.3 / 89.6 %	92.2 / 93.0 / 90.8 %	92.2 / 93.0 / 90.8 %
η at 25% $P_{ac,r}$[4)]	89.3 / 89.9 / 86.8 %	90.8 / 91.3 / 88.5 %	91.8 / 92.7 / 90.2 %	92.1 / 92.9 / 90.6 %	92.4 / 93.5 / 91.6 %	92.4 / 93.5 / 91.6 %
η at 30% $P_{ac,r}$[4)]	90.1 / 90.7 / 87.9 %	91.5 / 92.3 / 89.8 %	92.3 / 93.2 / 90.9 %	92.4 / 93.3 / 91.1 %	92.5 / 93.6 / 92.1 %	92.5 / 93.6 / 92.1 %
η at 50% $P_{ac,r}$[4)]	92.0 / 92.9 / 90.3 %	92.6 / 93.7 / 91.4 %	92.8 / 94.0 / 92.4 %	92.7 / 93.9 / 91.5 %	92.9 / 94.3 / 92.3 %	92.9 / 94.3 / 92.3 %
η at 75% $P_{ac,r}$[4)]	92.7 / 93.8 / 91.7 %	92.8 / 94.3 / 92.6 %	92.4 / 94.3 / 92.8 %	92.9 / 94.1 / 92.6 %	92.5 / 94.1 / 92.9 %	92.5 / 94.1 / 92.9 %
η at 100% $P_{ac,r}$[4)]	92.8 / 94.2 / 92.5 %	92.4 / 94.0 / 92.9 %	92.0 / 93.4 / 92.6 %	92.5 / 94.3 / 92.9 %	92.0 / 93.7 / 92.7 %	92.0 / 93.7 / 92.7 %
MPP adaptation efficiency	>99.9 %					

[4)] and at $U_{mpp\ min}$ / $U_{dc,r}$ / $U_{mpp\ max}$

PROTECTIVE EQUIPMENT	IG 15	IG 20	IG 30	IG 40	IG 50[1)]	IG 60 HV
DC insulation measurement	Warning/shutdown (depending on country setup) at R_{ISO} < 500 kOhm					
Overload behaviour	Operating point shift, power limitation					
Reverse polarity protection	Integrated					

be found at www.fronius.com.

Text and illustrations were accurate at the time of going to press. We reserve the right to make changes.
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